

```
1 #include <stdio.h>
2
3 #define MAX 10
4
5 int main()
6 {
7     int graph[MAX][MAX];
8     int vertices, edges;
9     int i, j, src, dest;
10
11     printf("Enter number of vertices: ");
12     scanf("%d", &vertices);
13
14     // Initialize matrix with 0
15     for(i = 0; i < vertices; i++)
16     {
17         for(j = 0; j < vertices; j++)
18         {
19             graph[i][j] = 0;
20         }
21     }
22 }
```

close ad [x]

input

```
Enter number of vertices: 2
Enter number of edges: 4
Enter edge (source destination): 5
```

```
23 printf("Enter number of edges: ");
24 scanf("%d", &edges);
25
26 // Input edges
27 for(i = 0; i < edges; i++)
28 {
29     printf("Enter edge (source destination): ");
30     scanf("%d %d", &src, &dest);
31
32     graph[src][dest] = 1;
33     graph[dest][src] = 1; // Undirected graph
34 }
35
36 // Display adjacency matrix
37 printf("\nAdjacency Matrix:\n");
38
39 for(i = 0; i < vertices; i++)
40 {
41     for(j = 0; j < vertices; j++)
42     {
43         printf("%d ", graph[i][j]);
44     }
```

input

```
Enter number of vertices: 2
Enter number of edges: 4
Enter edge (source destination): 5
```

```
30     scanf("%d %d", &src, &dest);
31
32     graph[src][dest] = 1;
33     graph[dest][src] = 1; // Undirected graph
34 }
35
36 // Display adjacency matrix
37 printf("\nAdjacency Matrix:\n");
38
39 for(i = 0; i < vertices; i++)
40 {
41     for(j = 0; j < vertices; j++)
42     {
43         printf("%d ", graph[i][j]);
44     }
45
46     printf("\n");
47 }
48
49 return 0;
50 }
51
```

close ad [x]



input

```
Enter number of vertices: 2
Enter number of edges: 4
Enter edge (source destination): 5
```

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 10
5
6 // Node structure
7 struct Node
8 {
9     int vertex;
10    struct Node *next;
11 };
12
13 struct Node *adjList[MAX];
14
15 // Create new node
16 struct Node* createNode(int v)
17 {
18     struct Node *newNode;
19
20     newNode = (struct Node*)malloc(sizeof(struct Node));
21
22     newNode->vertex = v;
23     newNode->next = NULL;
24
25     return newNode;
26 }
27
28 // Add edge
29 void addEdge(int src, int dest)
```

```
30 {
31     struct Node *newNode;
32
33     // Add edge src -> dest
34     newNode = createNode(dest);
35     newNode->next = adjList[src];
36     adjList[src] = newNode;
37
38     // Add edge dest -> src (Undirected graph)
39     newNode = createNode(src);
40     newNode->next = adjList[dest];
41     adjList[dest] = newNode;
42 }
43
44 // Display adjacency List
45 void display(int vertices)
46 {
47     int i;
48     struct Node *temp;
49
50     printf("\nAdjacency List:\n");
51
52     for(i = 0; i < vertices; i++)
53     {
54         printf("%d -> ", i);
55
56         temp = adjList[i];
57
58         while(temp != NULL)
```

main.c

```

57
58     while(temp != NULL)
59     {
60         printf("%d -> ", temp->vertex);
61         temp = temp->next;
62     }
63
64     printf("NULL\n");
65 }
66 }
67
68 int main()
69 {
70     int vertices, edges;
71     int i, src, dest;
72
73     printf("Enter number of vertices: ");
74     scanf("%d", &vertices);
75
76     // Initialize adjacency list
77     for(i = 0; i < vertices; i++)
78     {
79         adjList[i] = NULL;
80     }
81
82     printf("Enter number of edges: ");
83     scanf("%d", &edges);
84
85     // Input edges

```

plc cont

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```
78 {  
79     adjList[i] = NULL;  
80 }  
81  
82 printf("Enter number of edges: ");  
83 scanf("%d", &edges);  
84  
85 // Input edges  
86 for(i = 0; i < edges; i++)  
87 {  
88     printf("Enter edge (source destination): ");  
89     scanf("%d %d", &src, &dest);  
90  
91     addEdge(src, dest);  
92 }  
93  
94 display(vertices);  
95  
96 return 0;  
97 }  
98  
99
```



```
71 int i, src, dest;
72
73 printf("Enter number of vertices: ");
74 scanf("%d", &vertices);
75
76 // Initialize adjacency list
77 for(i = 0; i < vertices; i++)
78 {
79     adjList[i] = NULL;
80 }
81
82 printf("Enter number of edges: ");
83 scanf("%d", &edges);
84
85 // Input edges
86 for(i = 0; i < edges; i++)
87 {
88     printf("Enter edge (source destination): ");
89     scanf("%d %d", &src, &dest);
90
91     addEdge(src, dest);
92 }
93
94 display(vertices);
95
96 return 0;
97 }
```



```

78 {
79     adjList[i] = NULL;
80 }
81
82 printf("Enter number of edges: ");
83 scanf("%d", &edges);
84
85 // Input edges
86 for(i = 0; i < edges; i++)
87 {
88     printf("Enter edge (source destination): ");
89     scanf("%d %d", &src, &dest);
90
91     addEdge(src, dest);
92 }
93
94 display(vertices);
95
96 return 0;
97 }
98
99

```

input

```

Enter number of vertices: 5
Enter number of edges: 4
Enter edge (source destination): 2

```